

Analysis of Socioeconomic Factors Associated with County-Level Crime Rates in the United States

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A. Project Highlights

The goal of this project was to determine which socioeconomic factors were most strongly associated with violent crime rates and property crime rates across U.S. counties. Specifically, the analysis examined whether poverty rate, bachelor's degree attainment, median household income, unemployment rate, and metropolitan status were associated with county-level crime rates and whether those relationships differed between violent crime and property crime.

The project included collecting, cleaning, and merging county-level crime data from the FBI Crime Data Explorer with socioeconomic data from the U.S. Census Bureau American Community Survey (ACS) and unemployment data from the Bureau of Labor Statistics (BLS). The completed dataset was used to analyze the relationships between the selected socioeconomic variables and county-level violent and property crime rates. The project included descriptive statistics, exploratory data analysis, Pearson correlation analysis, and multiple linear regression. Deliverables included a cleaned analysis-ready dataset, Python scripts used for data preparation, a Jupyter Notebook documenting the analysis, visualizations, and this final report.

The completed solution was implemented using Python and Jupyter Notebook. Python libraries including pandas, NumPy, matplotlib, seaborn, scikit-learn, and Statsmodels were used to clean and prepare the data, perform statistical analyses, develop regression models, and create visualizations. The project followed the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology throughout the project to guide the data collection, preparation, analysis, and reporting processes.

B. Project Execution

The project was completed largely as originally planned. The project plan, objectives, and deliverables remained consistent throughout the project, and all planned deliverables were completed, including the cleaned dataset, Python data preparation scripts, Jupyter Notebook, statistical analyses, visualizations, and final report.

The project continued to follow the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology described in Task 2. The overall methodology did not change; however, one enhancement was made during implementation. The regression analysis was completed using the Statsmodels library in addition to scikit-learn so that regression coefficients, p-values, and additional model statistics could be reported more effectively. This change improved the statistical reporting while remaining consistent with the project's original analytical approach.

The project was completed within the originally planned timeline. Minor adjustments were made during data preparation after exploratory analysis identified several counties with extremely small populations that produced unstable per-capita crime rates. These counties were removed from the final analysis to improve the reliability of the regression models. Aside from this refinement and the addition of Statsmodels to provide more comprehensive regression statistics, the project milestones were completed as originally planned.

C. Data Collection Process

The data selection and collection process remained consistent with the original project plan. County-level crime data were obtained from the FBI Crime Data Explorer, socioeconomic data were obtained from the U.S. Census Bureau American Community Survey (ACS), and unemployment data were obtained from the Bureau of Labor Statistics (BLS). These datasets provided the variables required to address the research question and were successfully combined into a single analysis-ready dataset.

One challenge encountered during data collection was ensuring that the datasets could be merged accurately. Because the data originated from multiple government agencies, county Federal Information Processing Standards (FIPS) codes were standardized across all datasets before merging. During exploratory analysis, counties with extremely small populations were also identified as producing unstable per-capita crime rates. These counties were removed from the analysis to improve the stability and reliability of the statistical results.

No significant unplanned data governance issues were encountered during the project. All datasets were obtained from publicly available government sources, contained no personally identifiable information, and were appropriate for public use. Data quality was verified during preparation by checking for missing values, duplicate records, and successful dataset merges before beginning the statistical analysis.

C.1 Advantages and Limitations of Data Set

A major advantage of the dataset was that it combined reliable data from three authoritative U.S. government sources. Using county-level data provided broad geographic coverage and allowed socioeconomic

characteristics to be analyzed across more than 2,100 counties. The standardized reporting methods used by the FBI, Census Bureau, and Bureau of Labor Statistics also improved the consistency and reliability of the data.

One limitation of the dataset was that not all U.S. counties reported complete crime data to the FBI. Counties lacking sufficient crime reporting were excluded during data preparation, reducing the number of counties available for analysis. Additionally, the dataset included only a limited number of socioeconomic variables, meaning that other factors such as policing practices, demographic composition, income inequality, or housing conditions were not represented in the regression models and may also influence county crime rates.

D. Data Extraction and Preparation

Data extraction was performed by downloading publicly available county-level datasets from the FBI Crime Data Explorer, the U.S. Census Bureau American Community Survey (ACS), and the Bureau of Labor Statistics (BLS). These datasets were selected because they provided reliable, standardized measures for the crime and socioeconomic variables required to address the research question. Python and the pandas library were used to import the datasets into DataFrames for processing and analysis.

Data preparation consisted of several steps to ensure the datasets could be accurately combined and analyzed. County FIPS codes were standardized across all datasets to create a common identifier for merging. Only the variables required for the analysis were retained, and counties lacking sufficient FBI crime reporting were excluded. U.S. territories were also removed because they were outside the scope of the study. During exploratory analysis, counties with extremely small populations were identified as producing unstable per-capita crime rates and were removed.

from the final dataset to improve the stability of the statistical analysis. The cleaned datasets were then merged into a single analysis-ready dataset, and additional validation was performed by checking for missing values, duplicate records, and successful merges.

Python, pandas, NumPy, and Jupyter Notebook were appropriate tools for this project because they provided efficient methods for importing, cleaning, validating, merging, and organizing multiple large datasets. These tools also supported reproducible data preparation by documenting each processing step within Python scripts before the statistical analyses were performed.

E. Data Analysis Process

E.1 Data Analysis Methods

Several complementary analytical methods were used to evaluate the relationships between socioeconomic variables and county-level crime rates. Descriptive statistics were first calculated to summarize the distributions of the variables and identify their overall characteristics. Exploratory data analysis was then performed using histograms, boxplots, scatterplots, and a correlation matrix to visualize distributions, identify potential outliers, and examine relationships among variables.

Pearson correlation analysis was used to measure the strength and direction of the linear relationship between each socioeconomic variable and the violent crime rate and property crime rate. Multiple linear regression was then performed to evaluate the combined relationship between the independent variables and each crime category while controlling for the effects of the

remaining variables. Regression coefficients, p-values, and the coefficient of determination (R^2) were used to evaluate the statistical significance and explanatory power of the regression models. These analytical methods were appropriate because they directly addressed the research question by identifying and quantifying relationships between socioeconomic characteristics and county-level crime rates.

E.2 Advantages and Limitations of Tools and Techniques

Python and Jupyter Notebook provided a flexible and reproducible environment for conducting the analysis. Libraries including pandas and NumPy efficiently supported data cleaning, merging, and transformation, while matplotlib and seaborn enabled clear visualizations of the data. Scikit-learn and statsmodels provided reliable implementations of multiple linear regression and supported statistical evaluation through regression coefficients, p-values, and model performance metrics.

One limitation of these tools is that they require programming knowledge and careful validation to ensure that data are processed correctly. Statistical software also cannot determine whether analytical methods are appropriate for a given research question or whether omitted variables may influence the results. These limitations were addressed by carefully reviewing the data preparation process, verifying dataset merges, evaluating regression assumptions, and interpreting the statistical results within the context of the project.

E.3 Application of Analytical Methods

The analysis began by importing and cleaning the datasets using Python. County FIPS codes were standardized to allow accurate merging of the FBI, ACS, and BLS datasets. Counties lacking sufficient FBI crime reporting, U.S. territories, and counties with extremely small populations were removed during data preparation. The merged dataset was then validated by checking for missing values and duplicate records.

Descriptive statistics were calculated to summarize the variables included in the analysis. Histograms, boxplots, and scatterplots were reviewed to examine variable distributions, identify potential outliers, and evaluate the relationships between socioeconomic variables and crime rates. A Pearson correlation matrix and individual Pearson correlation analyses were then performed to quantify the strength and direction of the relationships between each predictor and the two crime rate variables.

Finally, separate multiple linear regression models were developed for violent crime rates and property crime rates. Scatterplots were reviewed to verify that the relationships between the predictors and the dependent variables were generally linear, and the correlation matrix was examined to evaluate potential multicollinearity among the independent variables. No severe multicollinearity was identified. Statistical significance was evaluated using an alpha level of 0.05, and regression coefficients, p-values, and R^2 values were used to interpret the results and evaluate the performance of the models.

F Data Analysis Results

F.1 Statistical Significance

The analytical methods used to evaluate the research question included Pearson correlation analysis and multiple linear regression. For both methods, the null hypothesis stated that no statistically significant relationship existed between the selected socioeconomic variables and county-level crime rates. Statistical significance was evaluated using an alpha level of 0.05.

Pearson correlation analysis identified statistically significant relationships between several socioeconomic variables and both violent crime rates and property crime rates. Multiple linear regression further evaluated these relationships while controlling for the effects of the remaining independent variables. Regression coefficients, p-values, and the coefficient of determination (R^2) were used to evaluate the models.

The violent crime regression model explained approximately 7.7% of the variation in county violent crime rates, while the property crime regression model explained approximately 5.5% of the variation in county property crime rates. Poverty rate and bachelor's degree attainment were statistically significant predictors in both regression models. Median household income and metropolitan status were statistically significant only in the property crime model, while unemployment rate was not statistically significant in either model. Because multiple predictors produced p-values below the 0.05 significance level, the null hypothesis was rejected for those variables, providing evidence that several socioeconomic characteristics were significantly associated with county-level crime rates.

F.2 Practical Significance

Although the regression models explained only a modest proportion of the variation in county crime rates, the results provide meaningful insights into the socioeconomic factors most consistently associated with crime. Poverty rate and educational attainment demonstrated statistically significant relationships across both crime categories, suggesting that these variables may be useful indicators when examining county-level crime patterns.

For example, policymakers, researchers, or local government agencies could use these findings to help prioritize community investments or identify counties that may benefit from additional educational, economic, or social support programs. While the models should not be used to predict crime rates independently, they provide a useful foundation for understanding how selected socioeconomic characteristics are associated with differences in county crime rates.

F.3 Overall Success

The project successfully addressed the original research question by identifying and quantifying the relationships between selected socioeconomic variables and county-level violent crime and property crime rates. The completed analysis produced a cleaned, integrated dataset, documented data preparation procedures, informative visualizations, Pearson correlation analyses, and multiple linear regression models that supported interpretation of the findings.

Although the regression models explained only a portion of the variation in county crime rates, the project successfully identified several statistically significant relationships that were consistent with the research objectives. The completed notebook and supporting Python scripts also provide a

reproducible analytical workflow that can be expanded in future research by incorporating additional socioeconomic or demographic variables to improve model performance.

G. Conclusion

G.1 Summary of Conclusions

This project examined the relationships between selected socioeconomic characteristics and county-level violent crime and property crime rates across the United States. The analysis identified poverty rate and bachelor's degree attainment as the most consistent predictors across both crime categories, while median household income and metropolitan status demonstrated more limited relationships after controlling for the remaining variables.

Although several statistically significant relationships were identified, the regression models explained only a modest proportion of the variation in county crime rates. These findings suggest that socioeconomic conditions are associated with crime patterns, but that additional demographic, economic, and community-level factors likely contribute to differences in county crime rates. Overall, the project successfully addressed the research question and demonstrated that multiple socioeconomic characteristics are significantly associated with county-level crime rates.

G.2 Effective Storytelling

The visualizations used throughout the analysis supported effective storytelling by presenting the data in a logical progression from exploration to

statistical analysis. Histograms and boxplots were used to illustrate the distributions of the variables and identify potential outliers. Scatterplots with regression trend lines visually demonstrated the relationships between individual socioeconomic variables and crime rates before formal statistical testing was performed. The correlation matrix summarized the relationships among all variables in a single visualization, while the regression output quantified those relationships after controlling for the remaining predictors.

Using Jupyter Notebook allowed the code, visualizations, statistical output, and written interpretations to be presented together in a single document. This structure enabled each stage of the analysis to build upon the previous one, making it easier to understand how the final conclusions were reached.

G.3 Recommended Courses of Action

Based on the findings of this analysis, two courses of action are recommended.

First, future analyses should incorporate additional socioeconomic, demographic, and community-level variables such as income inequality, policing resources, housing conditions, population density, and age distribution. Including these variables may improve the explanatory power of the regression models and provide a more comprehensive understanding of the factors associated with county crime rates.

Second, government agencies and researchers should continue using publicly available county-level socioeconomic data to identify long-term trends and monitor changes in crime patterns over time. Combining multiple authoritative data sources and regularly updating the analysis could support more informed policy decisions and resource allocation at the county level.

H. Panopto Presentation

Video Link:

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=77fd2776-38d9-4cd3-ae2a-b4790106db06>

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Appendix A

Python Source Code

The Python scripts used to acquire, clean, validate, merge, and prepare the datasets for analysis are included in the `python_scripts` folder submitted with this project.

Appendix B

Data Files

The original datasets obtained from the FBI Crime Data Explorer, the US Census Bureau American Community Survey (ACS), and the Bureau of Labor Statistics (BLS), along with the cleaned datasets produced during preprocessing, are included in the data_raw and data_clean folders submitted with this project.

Appendix C

Jupyter Notebook

The completed Jupyter Notebook, `Crime_Analysis_Notebook.ipynb`, containing the exploratory data analysis, visualizations, Pearson correlation analyses, regression models, statistical outputs, and written interpretations is included as a supporting file with this submission.